

Enrolment Number

Total No. of printed pages = 03

Winter, 2025

B. Tech 4th Semester Examination

Structural Analysis – Theory

Course Code: BCE23405T

Full Marks: 50

Time: 2 hours

The figure in the margin indicates full marks for the questions.

Section 1

Question 1

[10 Marks]

- a) What is the static indeterminacy of the beam fixed at both ends.
- b) In influence line diagrams (ILD) _____
 - a. Points remain fixed, position of load changes
 - b. Points change, position of load remains fixed
- c) If a unit rotation is applied at one end of the fixed beam, the moment produced at the other end of the beam will be _____
- d) Sway occurs when the portal frames are _____ in both geometry and loading.
- e) Which of the following is a statically determinate structure?
 - a. Two hinged arches
 - b. Three hinged arches
- f) If all the reaction on a planar system are concurrent in nature, then the system is _____
- g) The fixed end moment for a beam with uniformly distributed load (w kN/m) on entire span (L) is _____
- h) The strain energy stored in a beam due to axial force is given by _____
- i) The carry over factor for the prismatic beam is _____
- j) As per Muller Breslau principle, the influence line of a function is same as the _____ of the beam.

Section 2

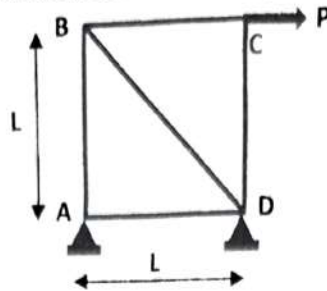
Answer any 4 questions [2 marks each]

Question 2

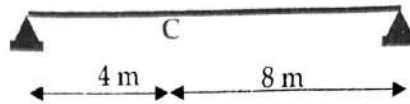
[8 Marks]

- a) What do you mean by principle of superposition. Write the equation of equilibrium for single plane structures.

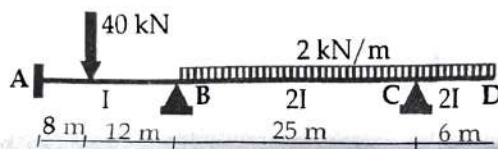
- b) Write the number of zero force members in the truss subjected to a force as shown in the figure



- c) A three hinged parabolic arch of span 20m and rise 4m carries a concentrated load of 150 kN at 4m from left support 'A'. Calculate the vertical reaction and the horizontal thrust respectively at support 'A'.
- d) Draw the influence line diagram for moment at point C of the given beam.



- e) Calculate the Distribution factor for each joint of the beam shown in figure below.



- f) A solid round bar 3 m long and 5 cm in diameter is used as a strut, determine the crippling load if one end hinged and another end fixed. Take $E = 2 \times 10^5 \text{ N/mm}^2$.

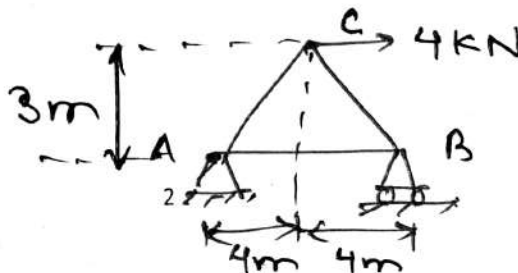
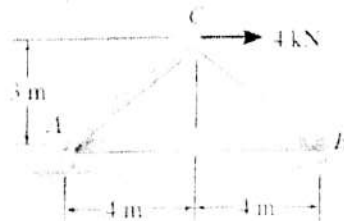
Section 3

Answer any 3 questions [4 marks each]

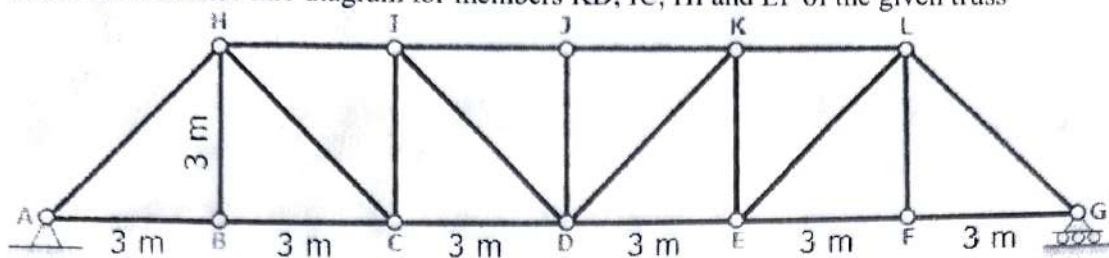
Question 3

[12 Marks]

- a) Derive the expression for crippling load for columns with both ends hinged.
- b) The cross-sectional area of each member of the truss shown in Figure is $A = 400 \text{ mm}^2$ and $E = 200 \text{ GPa}$. Determine the vertical displacement of joint C if a 4-kN force is applied to the truss at C. Use method of unit load.



- c) Draw the influence line diagram for members KD, IC, HI and EF of the given truss



- d) In the analysis of portal frame ABCD with sway, calculate the Restoring force (R) at the top of the frame.

The end column moments are as follows:

$$M_{AB} = 9.58 \text{ kN-m}$$

$$M_{CD} = -15.00 \text{ kN-m}$$

$$M_{BA} = 19.34 \text{ kN-m}$$

$$M_{DC} = 0 \text{ kN-m}$$

The length of column AB and CD are 10 m and 15 m respectively.

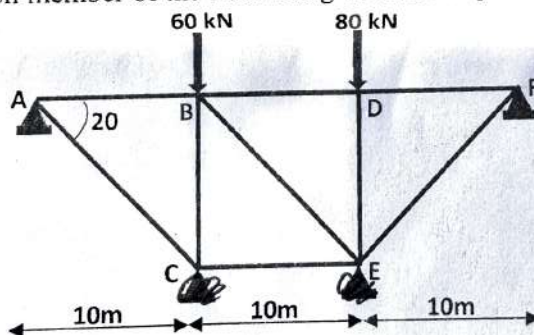
Section 4

Answer any 2 questions [10 marks each]

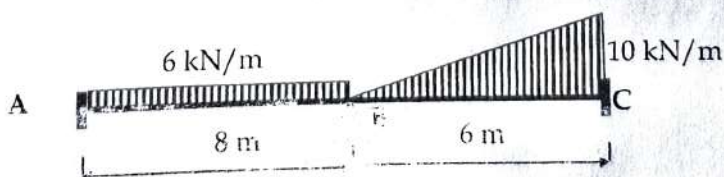
Question 4

[20 Marks]

- a) Calculate the forces in each member of the truss using **method of joints**.



- b) Analyse the beam shown in Figure using **Slope deflection method**. Draw the shear force and bending moment diagrams for the beam. Take EI as constant.



- c) Analyse the continuous beam shown in figure below using **Moment Distribution Method** if support B sinks by 10 mm. Draw Shear Force and Bending Moment Diagrams. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 4 \times 10^7 \text{ mm}^4$

